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Symmetric and Public keys

- DES stands for Data Encryption Standard.
- It is used to protect digital data.
- DES converts readable data (plaintext) into unreadable code (ciphertext).
- It uses a secret key for both encryption and decryption.
- It is a symmetric-key algorithm (same key is used to lock and unlock the data).
- DES processes data in 64-bit blocks.
- It uses a 56-bit key (with 8 extra bits for error checking).
- DES goes through 16 rounds of data transformation.

1. Initial Permutation (IP)

64-bit plaintext → Rearrange bits using IP table

2. Split into LPT and RPT

The permuted block is divided into two 32-bit halves:

1. LPT (Left Plain Text)
2. RPT (Right Plain Text)

These halves will be processed separately

3. 16 Rounds of Encryption

Each round expands, mixes with a subkey, substitutes (S-box), permutes, then swaps halves to scramble data.

4. Final Permutation (FP)

- After 16 rounds, the halves are combined
- Inverse of initial permutation

Example

Encrypt: "BYE" → IP → 16 Rounds → FP → "k8@Q"

Decrypt: "k8@Q" → FP^{-1} → 16 Rounds^{-1} → IP^{-1} → "BYE"

In information security, symmetric key encryption also known as symmetric key cryptography or secret-key encryption. Symmetric key is a method where the same secret key is used for both encrypting and decrypting data.

How Symmetric Key Works:

Key Generation: A single secret key is generated and must be securely shared between the communicating parties.

Encryption: The sender uses this key to convert plaintext (readable data) into ciphertext (unreadable data).

Transmission: The ciphertext is transmitted to the recipient over a communication channel.

Decryption: The recipient uses the same secret key to convert the ciphertext back into plaintext.

Example: Substitution Cipher (Caesar Cipher)

Scenario:

Ali wants to send the message “HELLO” to Bob using a Caesar cipher with a shift of 3.

Encryption Process:

Each letter in the plaintext is shifted three positions forward in the alphabet.

H → K

E → H

L → O

L → O

O → R

So, “HELLO” becomes “KHOOR”.

Decryption Process:

Bob, knowing the shift is 3, shifts each letter in “KHOOR” three positions backward:

K → H

H → E

O → L

O → L

R → O

Thus, he retrieves the original message “HELLO”.

- Uses a key pair: Public and Private.
- Public key for encryption, private key for decryption.
- Enables secure communication without key sharing.
- Algorithms: RSA
- It is also called as Asymmetric cryptography

Pros:

- - Solves key distribution issue
- - Scalable for many users

Cons:

- - Slower than symmetric algorithms
- - Computationally expensive

RSA example:

- Encryption:
 $C = M^e \bmod n$
- Decryption:
 $M = C^d \bmod n$

Step 1: Key Generation

- Bob generates:
 - **Public Key:** ($e = 3, n = 33$)
 - **Private Key:** ($d = 7, n = 33$)

Step 2: Encryption (Alice sends message to Bob)

- Alice wants to send number 4
- She uses Bob's **public key** to encrypt:

$$C = M^e \bmod n$$

$$C = 4^3 \bmod 33$$

$$C = 64 \bmod 33$$

$$C = 31$$

Step 3: Decryption (Bob receives and decrypts)

- Bob uses his **private key**:

$$M = C^d \bmod n$$

$$M = 31^7 \bmod 33$$

$$M = 4$$

Result: The original message 4 is recovered by Bob.

Symmetric Keys	Public Keys(Asymmetric)
Uses one single key for both encryption and decryption.	Uses two keys: a public key and a private key
The same key must be shared between sender and receiver.	Public key is shared with everyone while Private key is kept secret by the owner. What one key lock, the other can unlock.
Example: If I lock a box with a key, I must give you the same key to unlock it.	Example: You encrypt a message with my public key, only I can read it using my private key.
Algorithms: DES(Data Encryption Standard), AES(Advanced Encryption Standard)	Algorithms: RSA(Rivest–Shamir–Adleman), ECC(Elliptic Curve Cryptography)
Fast and efficient But sharing the key safely is difficult.	More secure for sharing But slower than symmetric encryption.

QUESTION?